Build-day checklist

Tick each box at the step it belongs to. mm throughout (cm in brackets where the build manual uses cm). Datum: top A's back-left corner (0, 0).

Measure before anything is cut, printed or drilled

- [ ] Both EKBACKEN tops 28 thick (calipers). If 27, shorten every screw into the top by 1 mm. Never deeper than 21 into a top. **MEASURE ON HARDWARE**
- [ ] Feet pad diameter 50 (calipers) before printing the 51-bore rings. **MEASURE ON HARDWARE**
- [ ] ALEX top panel at an edge: 16 or more for the 5/8 in ring screws. ALEX height 700 (with and without its stock feet). **MEASURE ON HARDWARE**
- [ ] Delivered rail lengths (bought as 84 in = 2133.6; layout 2130) and angle thickness 4.8. **MEASURE ON HARDWARE**
- [ ] VIVO lower clamp: maximum opening 31 or more (28 top + 3 plate); top arm reach (about 89). **MEASURE ON HARDWARE**
- [ ] Room: 252 × 317 cm of floor cleared, including the 30 wall gaps; wall to wall 2520 and 3170; baseboard thickness. **MEASURE ON HARDWARE**
- [ ] Feet load rating (design wants 50 kg or more each). **MEASURE ON HARDWARE**

Parts and tools on hand

- [ ] Six OLOV legs, 105 plate with five 5.5 holes on an 85 circle; 30 × M4 × 16 mm screws (H01), five per leg.
- [ ] Eight of the twelve 4 in feet: plate 59.5 with four 5.2 holes 46.8 apart; 32 × #8 × 5/8 in pan head (H02).
- [ ] Eight printed rings (51 bore, 3 wall, 10 ring on a 75 × 2.5 flange), PETG or ABS, flange-down; 24 × #6 × 5/8 in flat head (H06).
- [ ] Two rails, 2 × 2 × 3/16 in angle, ends deburred; 19 × #10 × 1 in pan head Robertson (H03).
- [ ] Four 10 × 3 in mending plates (two joining, two clamp pads); 8 × #10 × 5/8 in (H04) with #10 washers (H05).
- [ ] Two gusseted 90° strut fittings; 6 × 3/8-16 × 1-1/2 in grade 5 bolts (H08), 12 × 3/8 washers (H09), 6 nylon-insert nuts (H10).
- [ ] Printed tools: edge T-square, leg jig, feet corner jig, ALEX corner template, rail marking gauge, rail drill jig, two fitting centring plugs, drill guide (tower + 5 pucks), foot height gauge 102, wall gauge 65 / 30.
- [ ] Automatic centre punch, cutting oil; 5/64, 1/8, 7/32 and 7/16 in bits (7/16 cobalt); Robertson #2 bit on a 3 in extension.

Verify before: at each step

Step 1 · Steel on the bench

- [ ] Rail A holes at 3.5, 28, 54, 80, 106, 132, 158, 184, 210.5 cm from the wall end (first at 3.5, not 2.5); rail B at 2.5, 7.5, 28 … 210.5 cm from the corner end; all 19 from the flat leg's free edge.
- [ ] Fitting as template: rail A heel on the 52.0 cm mark legs toward the far end, and on 51.5 cm legs toward the wall end; rail B heel on the corner end. Holes 29 below the heel edge, 7/16 in.

Steps 2–3 · Marking the tops, face down

- [ ] Unfinished edge to the wall (y = 0 on top A, x = 0 on top B); the side-wall end of top A taped and measured from.
- [ ] A mark off by more than 2 mm: redraw it.

Step 4 · Legs, feet and rails

- [ ] Every OLOV pre-set to 802 (80.2 cm), or the ALEX-feet table's number (decided before this step).
- [ ] Every foot set with the 102 gauge: slides over from the side with no rock.
- [ ] Rail A: 120 (12 cm) from the front edge to the vertical leg's outer face at both ends and in the middle before drilling; end screws first, re-check, then the rest. Rail pilots 25 from the flat leg's top face (20 into the wood).
- [ ] Rail B: holes 3 to 10 only; holes 1 and 2 wait for step 7.

Step 5 · ALEX units and rings

- [ ] ALEX 1 back 65 off the back wall (wall gauge), near side 2125 (212.5 cm) from the side wall; ALEX 2 side 65 off the side wall, near end 2775 (277.5 cm) from the back wall.
- [ ] ALEX top panel 16 or more before the ring screws; ring centres 50 in from each ALEX corner.

Steps 6–7 · Stand the tops, make the joint

- [ ] Top A 30 off both walls, pads in their rings, level along x and y, all four pads bearing.
- [ ] Top B's legs 2–3 mm short before sliding rail B under top A; then wound up to flush.
- [ ] Before the plates: rail B's end touches rail A's vertical leg with no gap, and its outer face lines up with the 52.0 mark.
- [ ] Surfaces flush by the feet, never by packing under the joining plates.
- [ ] Before the bolts: surfaces flush and rail B touching rail A; inside fitting gusset-down. Six bolts 25 N·m.

Step 8 · Mount and trim

- [ ] Pad plates on their outlines (x 200–454, y 5–81; y 200–454, x 5–81); clamps finger-tight at x = 400 and y = 400 on Option 1 holes until the crossbar is on.
- [ ] G5 top edge 669 under the 673 limit; if the mount will not reach, lower the G5 (down to 120 still clears the ELLOVEN).
- [ ] Trim: gap measured at five or six points per wall; gap set to the smallest minus 1 mm; desk slid back until the 65 leg just fits at each ALEX.

Step 9 · Final checks (chapter J)

- [ ] 1 · Surface 830 ± 3 at all six corners (plus ALEX feet height if fitted).
- [ ] 2 · Level across the joint reads flat both ways; a fingertip finds no step.
- [ ] 3 · 20 kg (case of water + books) on top B's inside corner, 100 in from both edges: drop ≤ 2 on a straightedge from L1's line; returns, no creak, nothing moved. Never stand or sit on the desk.
- [ ] 4 · All eight pads bear: none spins free or rocks when turned by hand.
- [ ] 5 · Straightedge across the joint at the inside corner: step written down: _____ mm.
- [ ] 6 · Sideways push along each wall: no sway beyond a few mm, no top moving on its ALEX.
- [ ] 7 · No screw tip visible or felt on either laminate face.
- [ ] 8 · Knees clear; rail lower edge 751 from the floor, 120 back from the edge.
- [ ] 9 · Both ALEX fronts open fully without touching a foot or a leg.

Two weeks · date _____

- [ ] Retighten every screw.
- [ ] Straightedge at the joint: _____ mm. Settled? Retighten the six fitting bolts to 25 N·m (18 ft·lb).
- [ ] Each printed ring and its three screws: no crack (reprint in PETG if cracked), screws tight.

One month · date _____

- [ ] Straightedge at the joint: _____ mm. Settled? Retighten the six fitting bolts.